UKA TARSADIA UNIVERSITY

Asha M. Tarsadia Institute of Computer Science and Technology (AMTICS)

B.Tech Computer Science & Engineering (Cyber Security) • Academic Year 2026-27

CampusCare: Smart Campus Facility & Complaint Tracker

Course: Web Technologies with .NET (ASP.NET Core MVC, EF Core & MySQL)

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1. Problem Statement & Motivation

In collegiate institutions, physical equipment failures (lab projectors, split ACs, electrical wiring, LAN connectivity, plumbing breakdowns) are recorded manually in paper logbooks or informal chat channels. This creates acute operational bottlenecks across campus facilities:

- **70%+ Resolution Delays: Zero centralized dispatch or SLA tracking.**
- **Zero Transparency: Students and faculty cannot track repair status.**
- **Fragmented Records: Lost tickets, duplicate entries, no audit trail.**

2. Project Objectives

- **Centralized Online Portal:** Lodge verified tickets with location & category in < 60s.
- **Automated Lifecycle:** Manage progression: Open -> In Progress -> Resolved -> Closed.
- **Strict Data Integrity:** Server-side DataAnnotations + Client-side jQuery validation.
- **EF Core Code-First ORM:** Relational MySQL mapping with automatic pre-seeding.
- **Operational Analytics:** Live dashboard displaying resolution SLA rates and breakdown.

3. Production Technology Stack

ASP.NET Core FRAMEWORK C# Controller & DI Pipeline	**EF Core** ORM DbContext & Migrations
MySQL 8.0 DATABASE ACID Relational Storage	**Razor Views** UI ENGINE Tag Helpers & Binding
Bootstrap 5 DESIGN 12-Col Responsive Grid	**jQuery Unobtrusive** VALIDATION Client Validation Engine

4. Architecture & Request Pipeline

1. Browser Client (HTTP Request)
User submits ticket form with CSRF token via Bootstrap UI.

2. Routing & Anti-Forgery Filter
Route matching {controller}/{action} + validates CSRF token.

3. Controller & Model Binding
ComplaintsController validates input via ModelState.IsValid.

4. EF Core & MySQL Execution
ApplicationDbContext executes parameterized LINQ against MySQL.

5. State Management & Response
TempData["SuccessMessage"] passed to Razor with PRG redirect.

★ The 7 Core .NET Concepts

1. MVC Architectural Pattern — Separation

- Model: Complaint.cs & Category.cs define entities.
- View: Razor views (.cshtml) render UI via Tag Helpers.
- Controller: Handles DI, actions, & response pipeline.

```
public class ComplaintsController : Controller {
    private readonly ApplicationDbContext _context;
}
```

2. EF Core Code-First with MySQL — ORM

- Configured via Pomelo.EntityFrameworkCore.MySql.
- Auto-seeds 5 Campus Categories & sample tickets.

```
builder.Services.AddDbContext<ApplicationDbContext>(opt =>
    opt.UseMySql(connStr, ServerVersion.AutoDetect(connStr)));
```

3. Complete CRUD Operations — Lifecycle

- Create: Form POST saves new ticket with urgency.
- Read: Filtered dashboard lists tickets with status.
- Update: Edit action updates technician & remarks.
- Delete: Confirmation modal safely removes ticket.

```
public async Task<IActionResult> Create(Complaint model) {
    _context.Add(model); await _context.SaveChangesAsync();
}
```

4. Form Handling & Anti-Forgery — CSRF Guard

- Strongly-typed bindings: asp-for, asp-items, validation-for.
- Protected with [ValidateAntiForgeryToken] against CSRF.

```
<form asp-action="Create" method="post">
  @Html.AntiForgeryToken()
  <input asp-for="Title" class="form-control" />
</form>
```

5. Model Validation (Server & Client) — Integrity

- Server-side: [Required], [StringLength(100)], [EmailAddress].
- Controller check: if (!ModelState.IsValid) return View(model);
- Client-side: Real-time jQuery Unobtrusive validation.

```
[Required(ErrorMessage = "Title is required")]
[StringLength(100)] public string Title { get; set; }
```

6. State Management (PRG Pattern) — TempData

- Post-Redirect-Get pattern with TempData["SuccessMessage"].
- Shows dismissible green toast alert after redirect.

```
TempData["SuccessMessage"] = $"Grievance #{model.Id} registered!";
return RedirectToAction(nameof(Index));
```

7. Responsive UI Design — Bootstrap 5

- Mobile-first 12-column grid collapses cleanly on phones.
- Features 4 KPI summary cards, filter bars, & status badges.

```
<div class="card border-0 shadow-sm rounded-4 h-100 p-3">...</div>
```

6. Implemented Application Modules

- **Student Issue Desk (Create)**
  Lodge grievances with Category, Urgency level, Location, and Email.
- **Admin Control Center (Index & Edit)**
  Filter by department, assign technician, update status, and add remarks.
- **SLA & Resolution Analytics**
  High-level summary displaying overall resolution rate & category breakdown.
- **Safe Removal & Deletion (Delete)**
  Two-step confirmation modal preventing accidental ticket elimination.

7. Live Application UI Walkthrough

http://localhost:5000/Complaints (Live Running Application)

02 OPEN TICKETS	01 IN PROGRESS	02 RESOLVED

ID	Grievance Title	Location	Priority	Status
#1	Projector HDMI Port ...	Seminar 204	High	In Progress
#2	Split AC Leaking Wat...	Lab 305	Urgent	Open
#3	Tube Lights Flickeri...	Library 1F	Low	Resolved
#4	Restroom Tap Broken...	Hostel 3	High	Open

8. Verification & Test Case Matrix

TC-01: Empty Form Submission Triggers DataAnnotations; blocks commit	PASS
TC-02: Valid Ticket Creation Persists to MySQL; triggers TempData toast	PASS
TC-03: Lifecycle Progression Open -> In Progress -> Resolved advance	PASS
TC-04: Anti-CSRF Guard Blocks unauthorized external POST requests	PASS
TC-05: Mobile Layout Rendering Bootstrap 12-col grid collapses gracefully	PASS
TC-06: Pre-Seeded Startup DbInitializer seeds categories & tickets	PASS

9. Conclusion & Future Enhancements

CampusCare fully satisfies the academic evaluation criteria: clean MVC separation, resilient EF Core database connectivity, robust server & client validation, TempData state management, and responsive presentation.

Planned Future Scope:

- Institutional Single Sign-On (SSO) via ASP.NET Core Identity.
- Automated Email/SMS alerts on ticket status changes via SendGrid.
- Automated maintenance dispatch via smart campus IoT sensors.